WEEK 1

The Broken Toy

Session 01 / The cantilever beam

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The QUESTION

Which coordinates expose the structure of the task?

The projectile challenged the model’s shape.
The cantilever challenges its coordinate system.

What are these?

64	68	55	70	76	64	67	74	70	70	76	81	84	89	82	74
60	67	148	158	151	95	73	77	71	72	76	74	74	138	77	69
65	104	173	148	174	133	75	77	71	82	91	92	101	136	132	130
66	73	124	169	135	146	85	89	96	117	131	143	149	171	162	161
81	77	80	114	144	181	208	194	194	201	200	197	181	192	158	156
112	113	120	105	112	176	202	202	206	192	154	157	196	176	138	136
140	142	140	145	133	168	191	183	171	153	129	152	180	177	136	138
139	147	136	164	152	163	176	154	135	115	68	111	184	168	115	114
137	134	127	137	85	117	145	114	102	117	75	77	180	170	124	117
106	112	113	112	111	93	166	135	125	124	111	85	144	164	115	117
114	113	113	115	133	111	158	135	127	129	126	97	115	129	118	118
130	132	114	121	135	122	152	136	132	123	122	125	112	134	122	120
114	116	122	131	137	125	148	135	130	121	116	111	117	125	124	109
109	108	115	137	135	113	164	113	110	105	105	100	117	134	131	108
116	115	112	118	96	102	143	104	110	108	103	104	102	103	109	114
116	118	114	115	114	114	111	116	106	112	108	120	124	121	119	115

179	128	83	95	75	55	61	106	121	97	51	55	54	52	40	76
121	109	77	62	40	54	78	139	175	171	43	48	51	48	38	54
172	160	136	145	159	162	155	147	136	151	126	122	120	122	117	112
149	152	151	158	157	157	155	136	108	141	157	153	150	150	147	142
143	146	142	142	154	103	64	113	105	132	101	139	138	137	138	137
141	141	141	143	133	41	55	101	142	140	62	88	136	136	133	130
142	134	142	144	112	28	51	77	114	123	53	71	126	123	121	127
130	126	127	136	85	17	32	43	118	76	33	70	135	122	117	118
126	118	122	139	66	18	28	27	106	31	18	63	135	132	125	113
125	117	115	122	39	19	28	22	39	23	39	64	176	187	114	119
130	116	114	111	35	26	28	21	21	26	68	62	186	176	108	104
119	114	107	103	32	27	27	26	23	37	76	64	186	163	145	105
117	106	97	99	123	35	40	45	29	55	97	90	155	136	110	104
111	93	93	93	84	33	33	45	31	69	107	116	135	125	101	101
99	95	95	82	79	36	30	34	48	79	96	103	128	118	93	89
97	89	88	88	82	37	61	147	96	88	82	66	139	150	94	88

Two grids of numbers. Same shape, same range, same statistics.

Same structures. Different presentation.





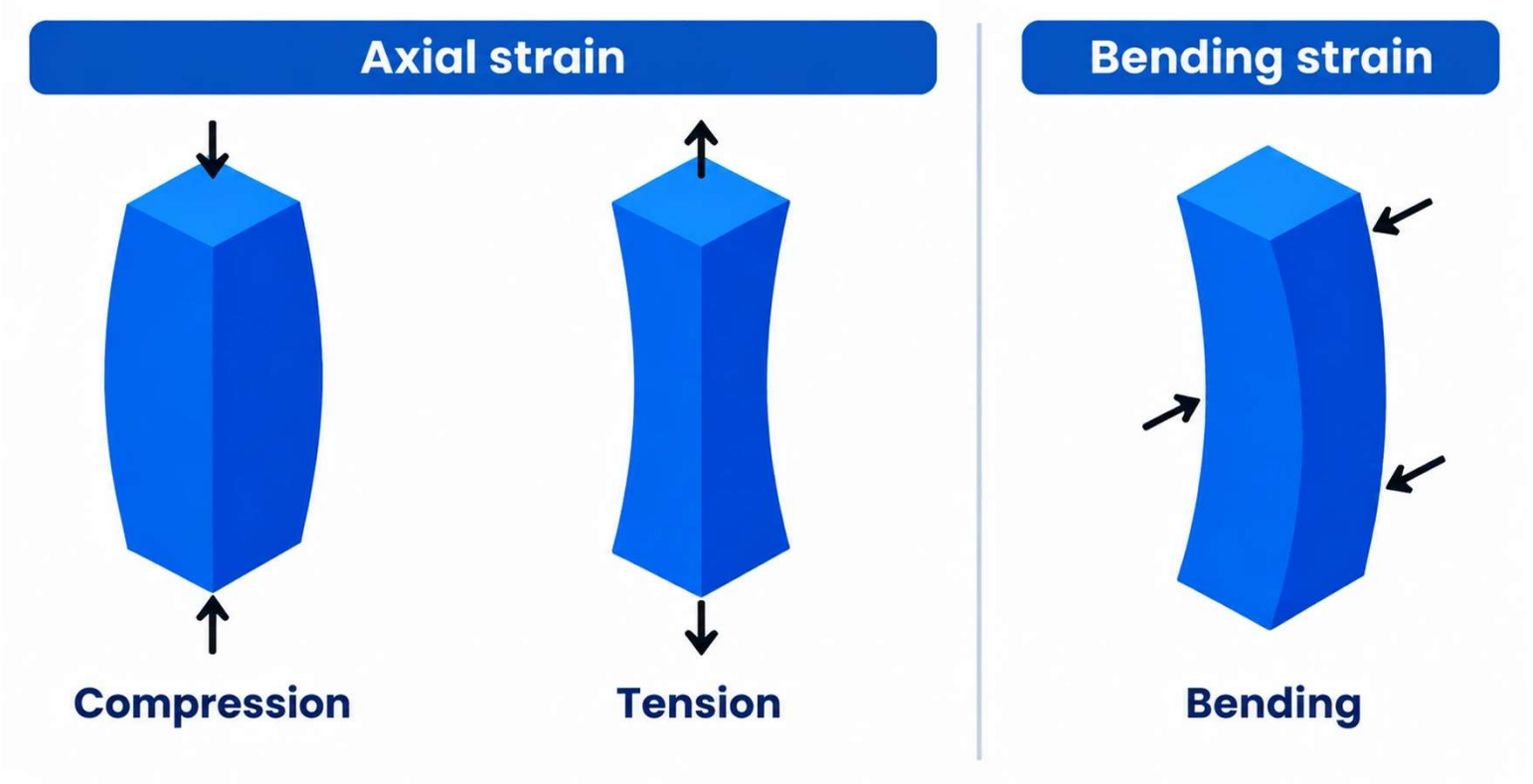
Edges and regions make the arrangement visible.

PART 2

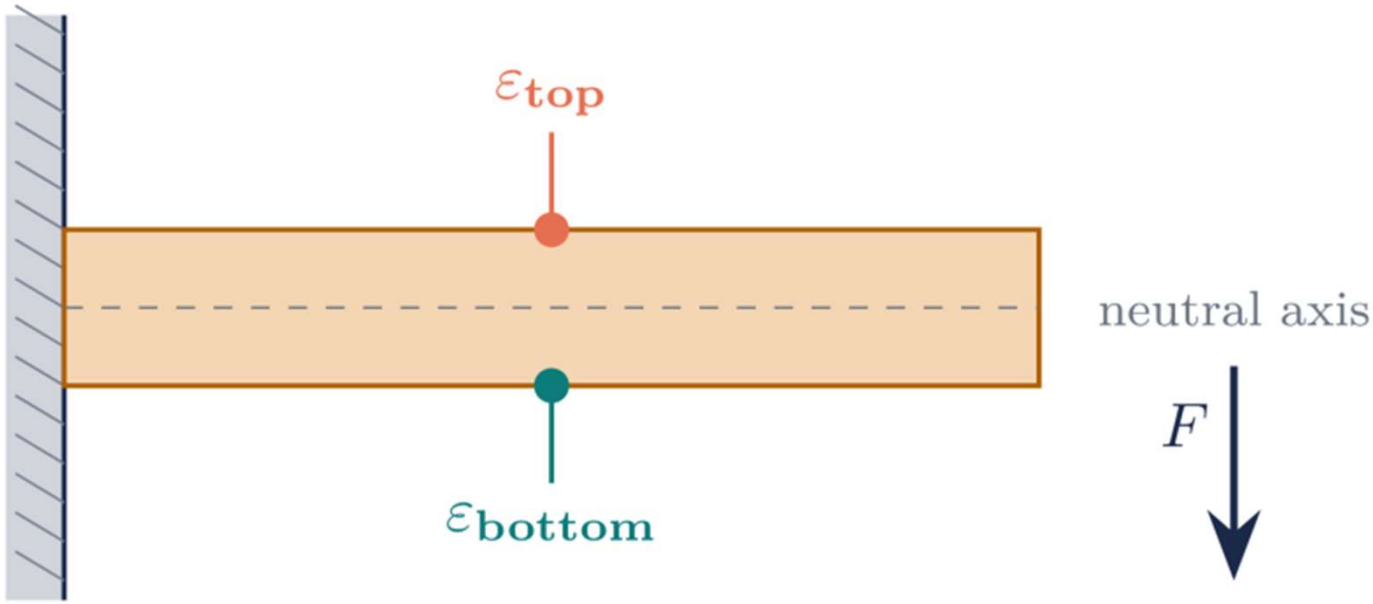
Representation is not decoration. It is the difference between meaning and noise.

Now we will see the same lesson in a physical system, where we cannot cheat with color.

Axial vs bending strain

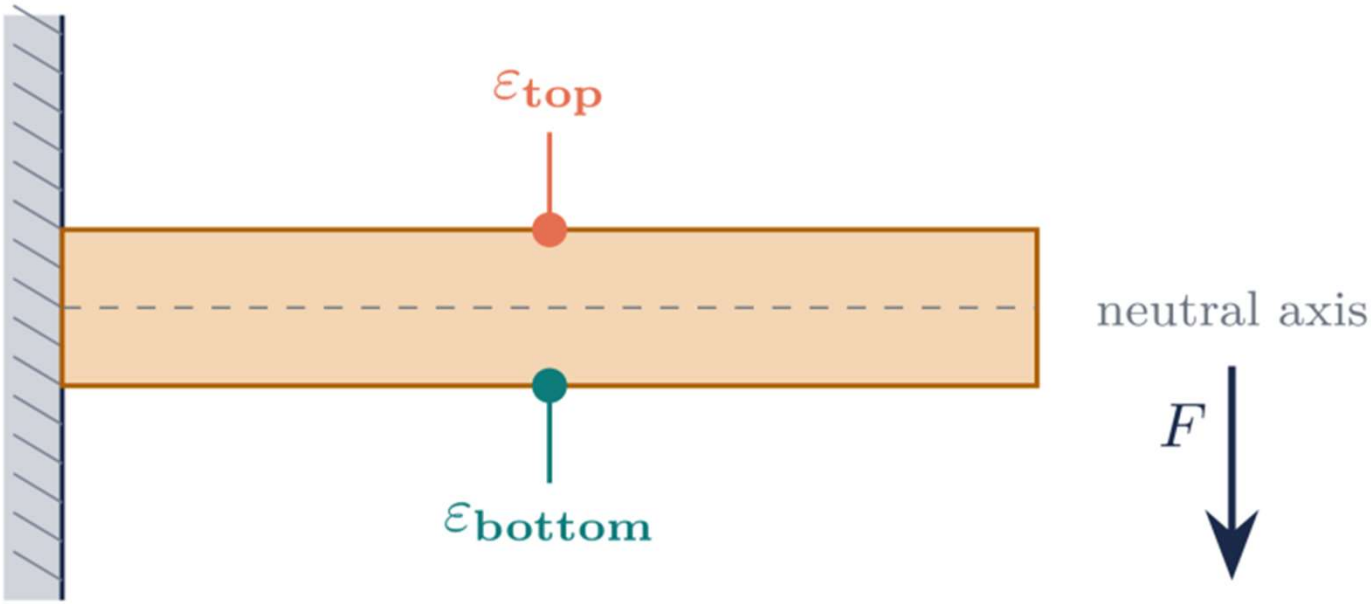


A cantilever with two strain gauges



A cantilever beam with two strain gauges at the same cross-section

A cantilever with two strain gauges



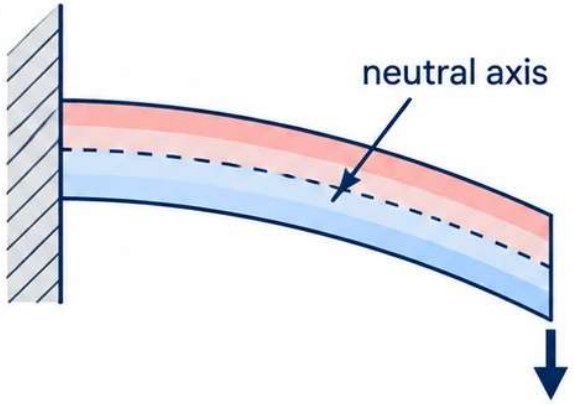
A cantilever beam with two strain gauges at the same cross-section

The beam can experience two different states

- **Axial strain ε_a**: tension or compression.
  Same sign on top and bottom.



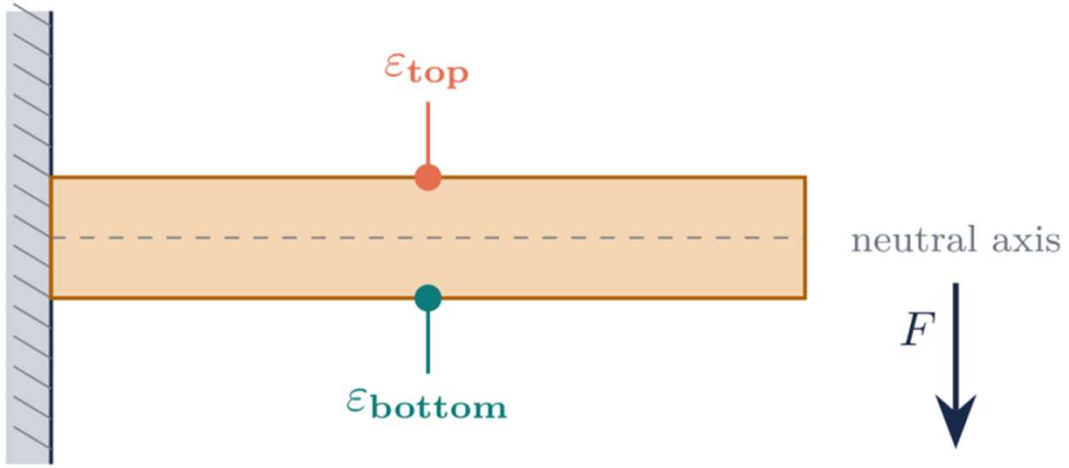
- **Bending strain ε_b**: bending.
  Opposite signs on top and bottom.



The beam can experience two different states

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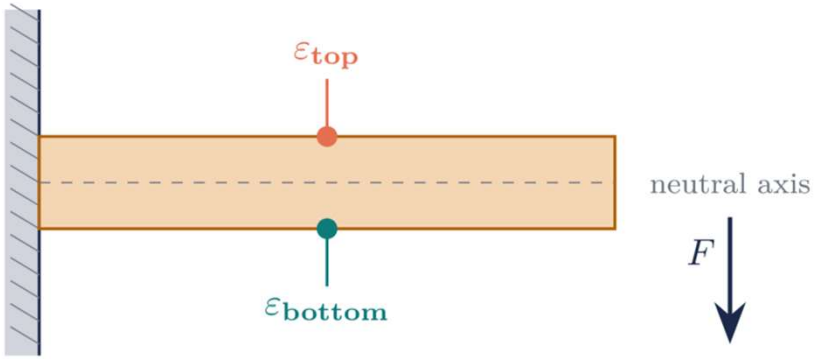
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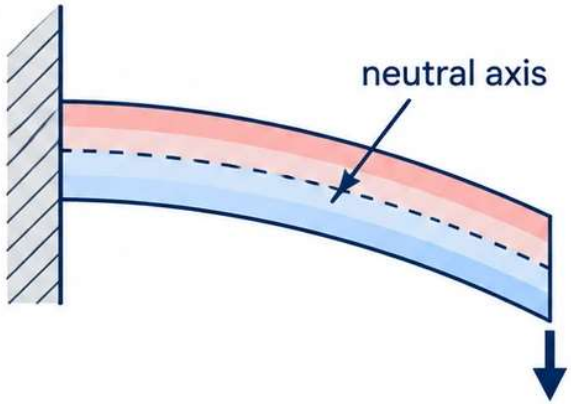
$$\varepsilon_{top} = \varepsilon_a$$

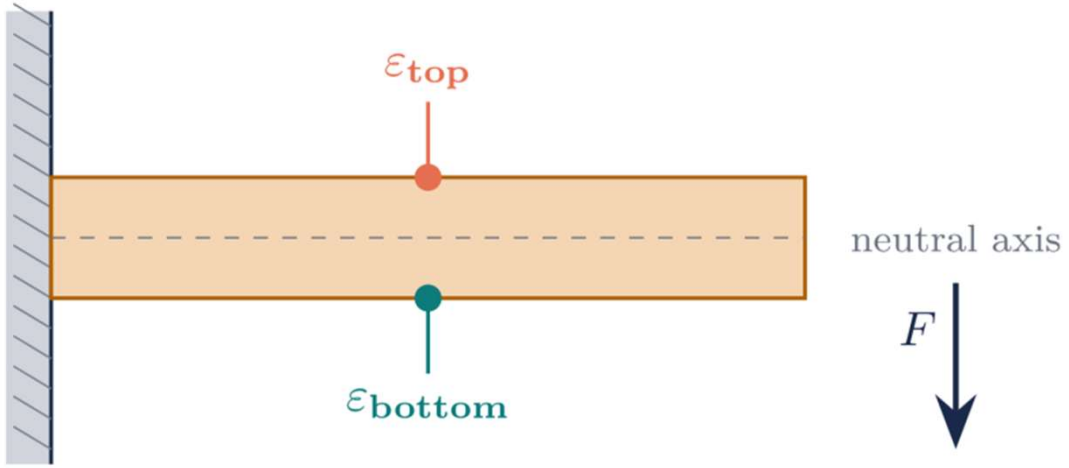
$$\varepsilon_{bottom} = \varepsilon_a$$



The beam can experience two different states

- **Bending strain ε_b**: bending.
  Opposite signs on top and bottom.





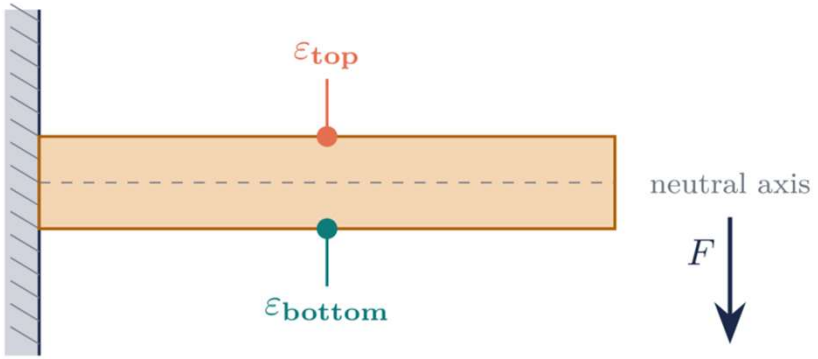
The beam can experience two different states

- **Axial strain ε_a**: tension or compression.
  Same sign on top and bottom.

$$\varepsilon_{top} = \varepsilon_a$$
$$\varepsilon_{bottom} = \varepsilon_a$$

- **Bending strain ε_b**: bending.
  Opposite signs on top and bottom.

$$\varepsilon_{top} = \varepsilon_b$$
$$\varepsilon_{bottom} = -\varepsilon_b$$



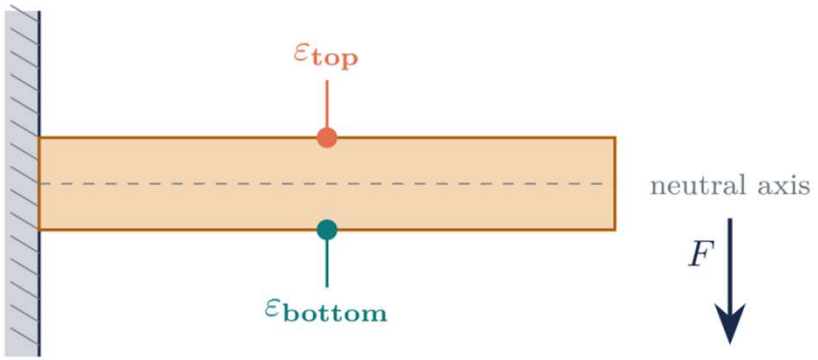
The observation mixes both causes

$$\varepsilon_{top} = \varepsilon_a + \varepsilon_b + \eta_{top}$$

$$\varepsilon_{bottom} = \varepsilon_a - \varepsilon_b + \eta_{bottom}$$

$$\varepsilon_{top} = \varepsilon_a$$
$$\varepsilon_{bottom} = \varepsilon_a$$

$$\varepsilon_{top} = \varepsilon_b$$
$$\varepsilon_{bottom} = -\varepsilon_b$$



The task: detecting positive bending

Target	$y = 1$ when the true bending component $\varepsilon_b > 0$.
Boundary	$\varepsilon_{top} = \varepsilon_{bottom}$ is a diagonal in the plane.
Model family	A shallow decision tree splits one coordinate at a time.

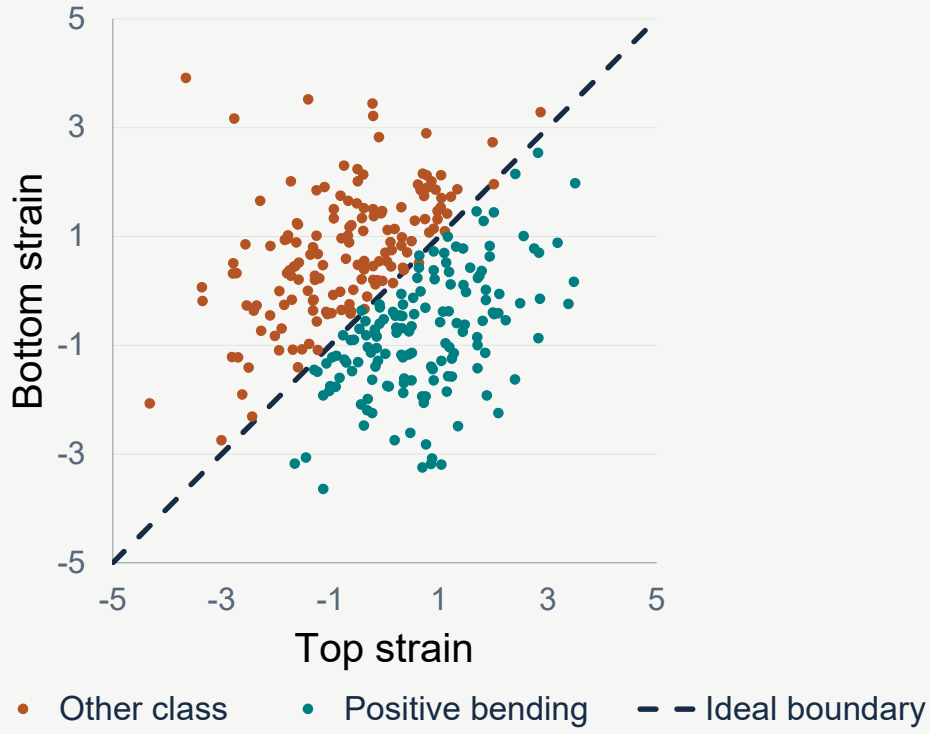
The raw coordinate plane

PART 3

Let’s try it

We’ll simulate this by python.

The raw coordinate plane

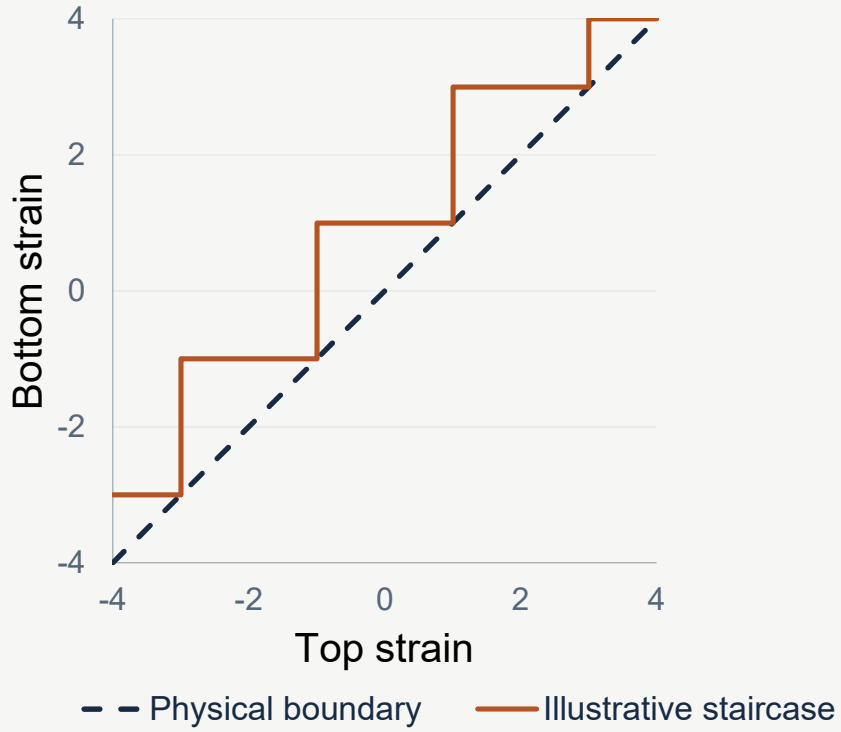


A diagonal boundary

Each raw axis mixes axial strain and bending.

The target depends on their difference.

Axis-aligned splits form a staircase



More partitions

A tree approximates the diagonal with horizontal and vertical steps.

This curve illustrates the geometry.

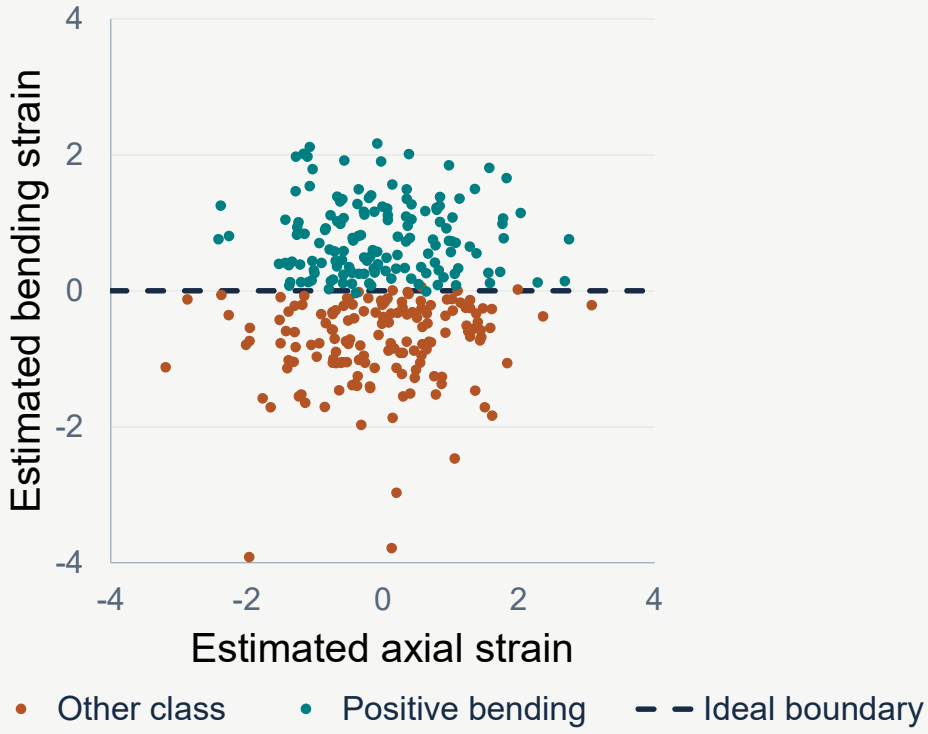
THE REPRESENTATIONAL SHIFT

$$axial = \frac{(\varepsilon_{top} + \varepsilon_{bottom})}{2}$$

$$bending = \frac{(\varepsilon_{top} - \varepsilon_{bottom})}{2}$$

Mechanics supplies the coordinates.

The physical coordinate plane



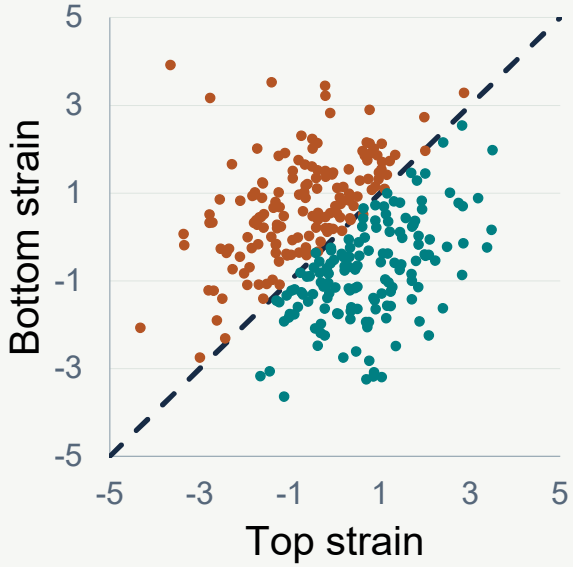
A single threshold

The decision now depends on one coordinate.

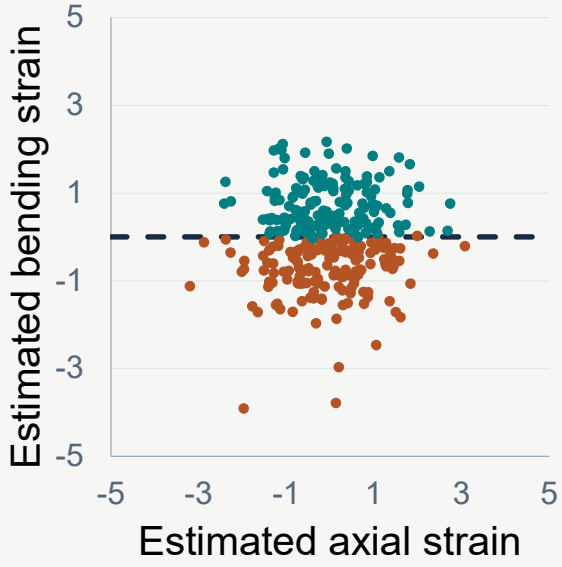
A split near bending = 0 matches the task.

The same samples in two bases

Raw coordinates



Physical coordinates



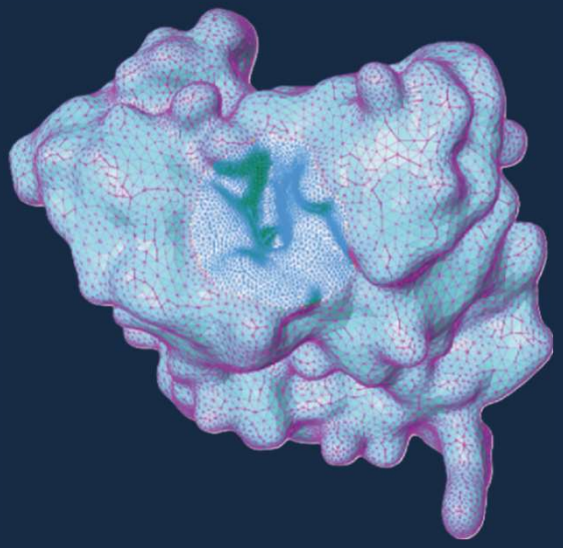
Other class Positive bending Ideal boundary

Other class Positive bending Ideal boundary

Same observations. Same labels. A simpler decision boundary.

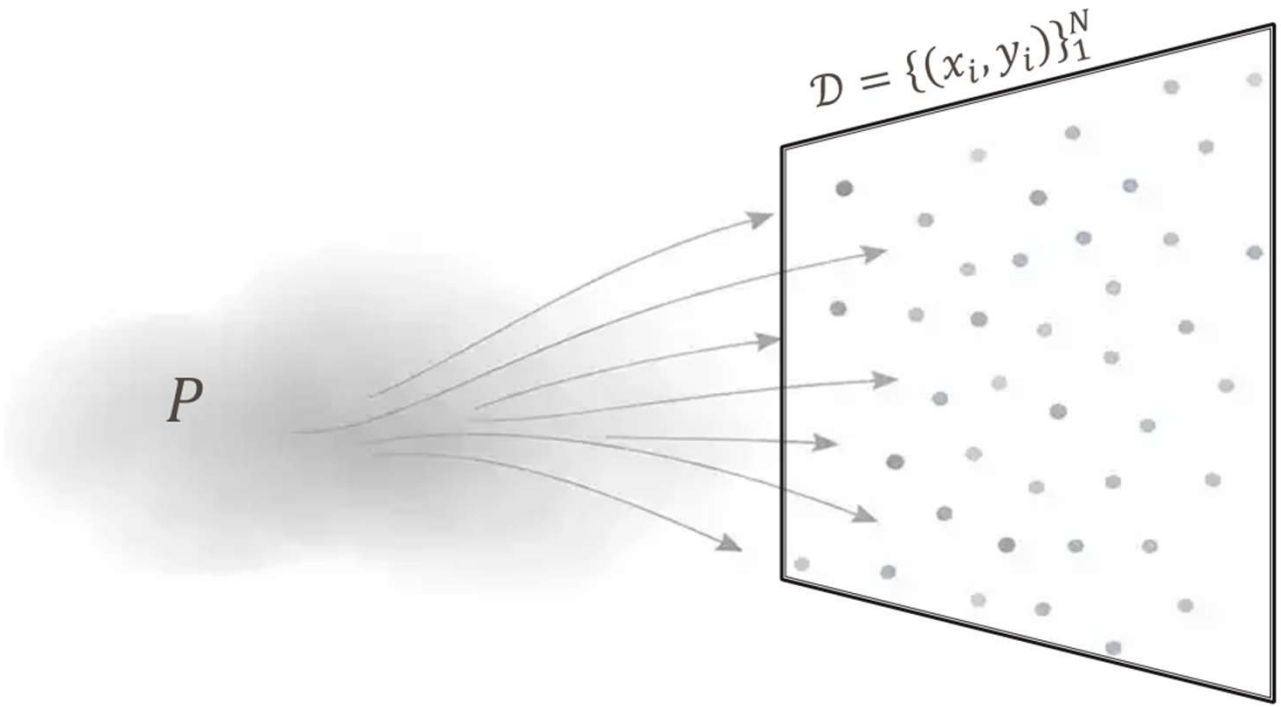
PART 4

From the Physical World to Raw Data



An underlying structure.

The physical world and raw data

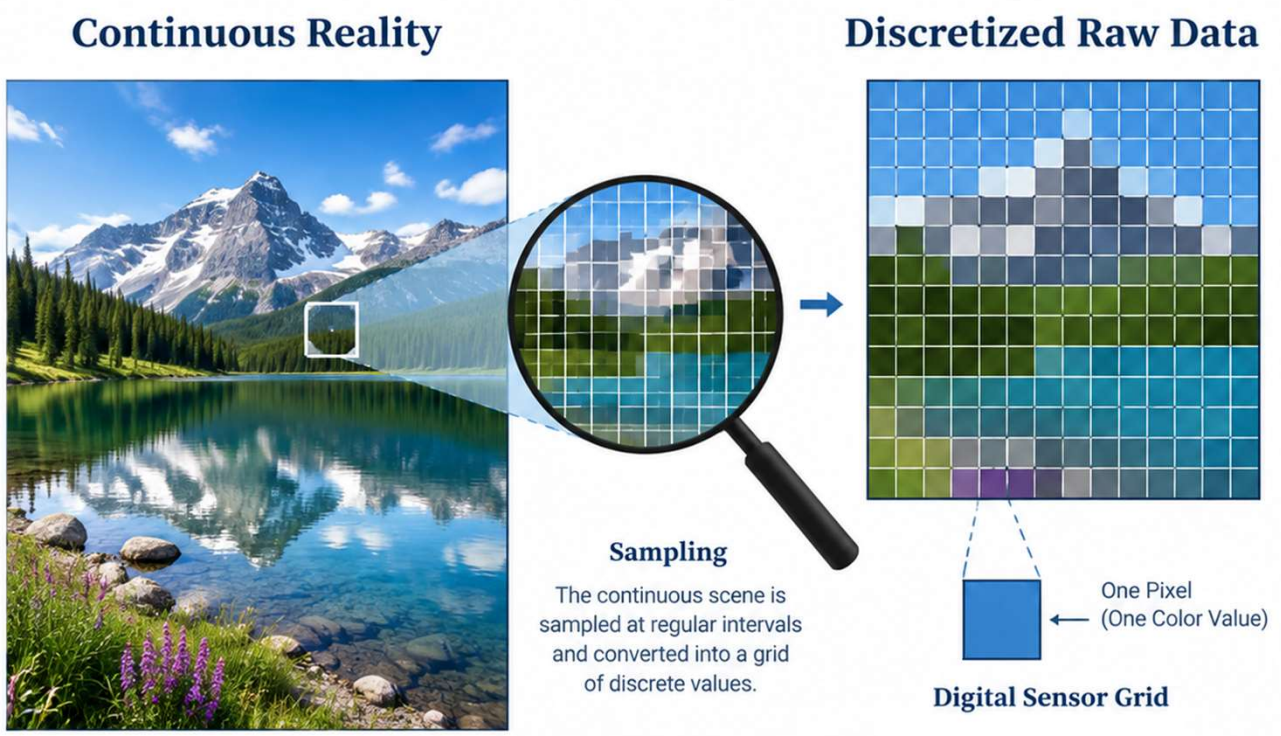


Finite observations

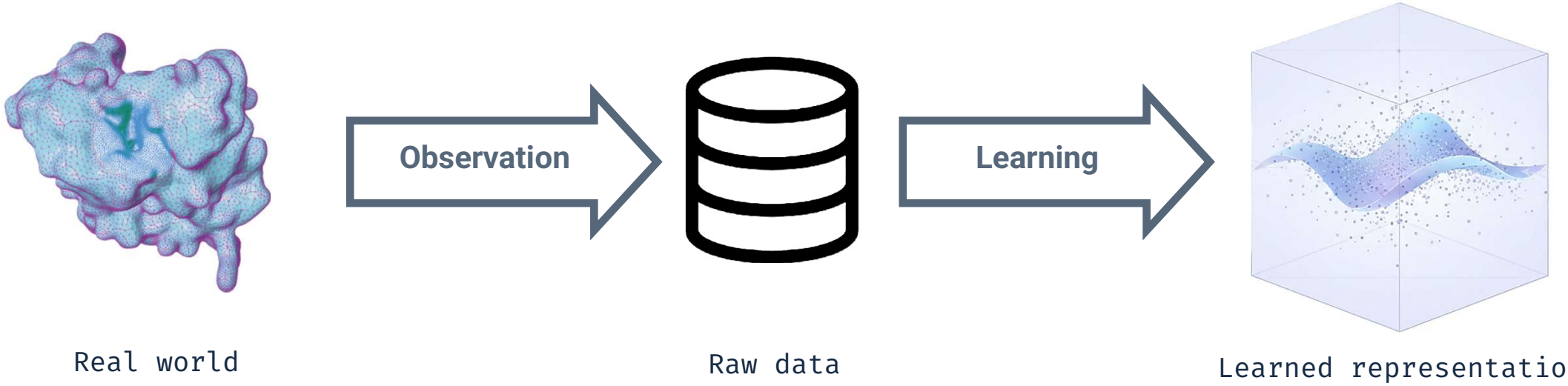
Sensors sample a richer phenomenon.

Precision, hidden variables and noise limit what we observe.

Sampling a continuous scene



Observation and representation



Two maps. Two chances to lose what matters.

The representation problem

The representation problem. Given a phenomenon P with structure $\mathcal{S}$ and a task T, an observation map $o: P \to \mathbb{R}^n$, find a representation $\phi: \mathbb{R}^n \to \mathbb{R}^m$ such that the structure relevant to T is preserved or made explicit in the coordinates $\phi\big(o(P)\big)$.

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The representation ϕ is the object of study. The algorithm $\mathcal{A}$ is the procedure that finds ϕ.

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The representation ϕ is the object of study. The algorithm $\mathcal{A}$ is the procedure that finds ϕ.

The rest of the course is the systematic study of $\mathcal{A}$.

Code: generating the measurements

```
import numpy as np

rng = np.random.default_rng(0)
n = 300
axial = rng.normal(0, 1.0, n)
bending = rng.normal(0, 1.0, n)
noise = 0.05 * rng.normal(0, 1, (n, 2))

top = axial + bending + noise[:, 0]
bottom = axial - bending + noise[:, 1]
```

Independent physical components mix at the sensors.

Code: the two representations

```
axial_c = (top + bottom) / 2
bending_c = (top - bottom) / 2

X_raw = np.c_[top, bottom]
X_phys = np.c_[axial_c, bending_c]
y = (bending > 0).astype(int)
```

The observations and labels stay fixed.
Only the input coordinates change.

Code: a controlled model comparison

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import cross_val_score

clf = DecisionTreeClassifier(
    max_depth=3, random_state=0
)

score_raw = cross_val_score(clf, X_raw, y, cv=5).mean()
score_phys = cross_val_score(clf, X_phys, y, cv=5).mean()
```

One classifier family and one capacity limit in both cases.

Illustrative results in the lecture note

Representation	Quoted accuracy
Raw coordinates	0.72
Physical coordinates	**0.99**

The note labels these values “Typical output”.

The geometric lesson concerns the ease of expressing the boundary.

Diagnosing a representation mismatch

1 Physical structure	Which causes, symmetries or relationships matter?
2 Observation	What does the sensor preserve, mix or discard?
3 Model bias	Can the model express the task efficiently in these coordinates?

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Here, the physical basis aligns the boundary with a tree split.

The same lesson across two sessions

Dimension	Projectile, Session 00	Cantilever, Session 01
Raw coordinates	Launch angle θ	Top and bottom strains
Failure mode	Chosen model misses shape	Coordinates mix causes
Fix	Change the features	Change the basis
Prior knowledge	Projectile mechanics	Beam deformation

Physical knowledge guided both representations.

SESSION 02: THE HIDDEN SHAPE

What if we do not know the physical coordinates?

Can the data reveal a useful representation?

Exercises 1–2: inversion and noise

1 The inverse observation map

Derive A^{-1}. Recover axial and bending components exactly from top and bottom readings when $\eta = 0$.

2 Removing measurement noise

Set $\eta = 0$. Why can a shallow tree still struggle with the raw coordinate plane?

Exercise 3: changing the model family

A shallow tree and logistic regression have different geometric biases.

Replace the tree with logistic regression.

Does the accuracy gap shrink, remain, or grow?

Explain the result using the decision boundary.

SESSION 01

The coordinates shape what a model can express.

The cantilever separates causes before classification.